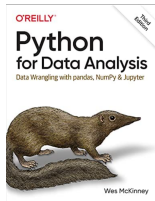


7.2: Data Wrangling and Cleaning

- **Instructor:** Dr. GP Saggese, gsaggese@umd.edu
- **Reference**
 - [Pandas tutorial](#)
 - Web
 - Onslaught of free resources (e.g., [ref](#))
 - Mastery
 - <https://wesmckinney.com/book>
 - Read cover-to-cover and try all the examples 2-3x time to really *master*



Data Wrangling and Data Cleaning

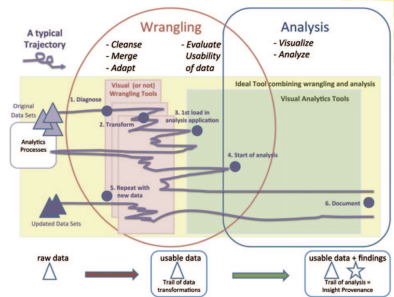
- **Data wrangling and cleaning**

- Turn *messy, inconsistent, incomplete* data into a *coherent* dataset

- **Most effort** in data science is spent here, not modeling

- It is a iterative work
- Clean → Modeling → Clean →

...



- **Data wrangling**

- Transform data into a useful structure
- Reshape tables, merge sources, create new variables

- **Data cleaning**

- Correct or remove incorrect, corrupt, or irrelevant data
- Handle missing values, outliers, inconsistent formats

- **Often done together**

Typical Data Wrangling Workflow

- **Inspect data**

- Look at column names, data types
- Look at a few rows
- Basic summaries

- **Diagnose problems**

- Missing values, weird codes, inconsistent units, strange distributions

- **Decide strategy**

- What to keep
- What to fix
- What to drop based on your question

- **Apply transformations**

- Filter, join, reshape, recode, aggregate

- **Validate results**

- Check that changes make sense and key statistics are reasonable



Data Extraction

- **Web scraping**
 - Use APIs or scrape data explicitly
 - Scraping is challenging
 - Fragile
 - Throttling
 - Cat-and-mouse with websites
 - Set up pipelines for periodic scraping
 - Tools for automated scraping
 - E.g., `import.io`, `portia`, ...
- **Various sources of data**
 - Files (e.g., CSV, JSON, XML)
 - Databases
 - Spreadsheets
 - AWS S3 buckets
 - Web APIs
 - Logs
 - ...
- Raw data reflects **collection method, not analysis needs**

Tidy Data

- **Definition:** Tidy data is a structured format for datasets that makes them easier to manipulate, analyze, and visualize
 - [Tidy data, Wickham, 2014](#)
- **Core Principles**
 - Each variable forms a *column*
 - Each observation forms a *row*
 - Each type of observational unit forms a *table*
 - `pandas.melt()`, `pandas.pivot()`

	treatmenta	treatmentb
John Smith	—	2
Jane Doe	16	11
Mary Johnson	3	1

	John Smith	Jane Doe	Mary Johnson
treatmenta	—	16	3
treatmentb	2	11	1

name	trt	result
John Smith	a	—
Jane Doe	a	16
Mary Johnson	a	3
John Smith	b	2
Jane Doe	b	11
Mary Johnson	b	1

Tidy data

“Messy” data

Reshaping data

- **Wide to long**
 - Turn similar columns into key-value pairs
- **Long to wide**
 - Turn key-value pairs into separate columns
- **What is the best shape?**
 - Choose shape for simple analysis and visualization

	type	date	clicks	conversions	impressions
0		2020-01-01	1.0	NaN	18.0
1		2020-01-02	2.0	NaN	19.0
2		2020-01-03	1.0	1.0	14.0
3		2020-01-04	NaN	NaN	5.0
4		2020-01-05	1.0	NaN	8.0
5		2020-01-06	1.0	1.0	15.0
6		2020-01-07	2.0	NaN	8.0

Wide format

	date	type	count
0	2020-01-01	impressions	18.0
1	2020-01-02	impressions	19.0
2	2020-01-03	impressions	14.0
3	2020-01-04	impressions	5.0
4	2020-01-05	impressions	8.0
...	...	...	...
91	2020-01-28	conversions	NaN
92	2020-01-29	conversions	NaN
93	2020-01-30	conversions	NaN
94	2020-01-31	conversions	NaN
95	2020-02-01	conversions	NaN

Long format

Split-Apply-Combine

- **Common data analysis pattern**

- *Split*: break data into smaller pieces
- *Apply*: operate on each piece independently
- *Combine*: combine pieces back together
- [The Split-Apply-Combine Strategy for Data Analysis, Wickam, 2011](#)

- **Examples**

- Group-wise ranking
- Group vars (sums, means, counts)
- Create new models per group

- **Pros**

- Code is compact
- Easy to parallelize

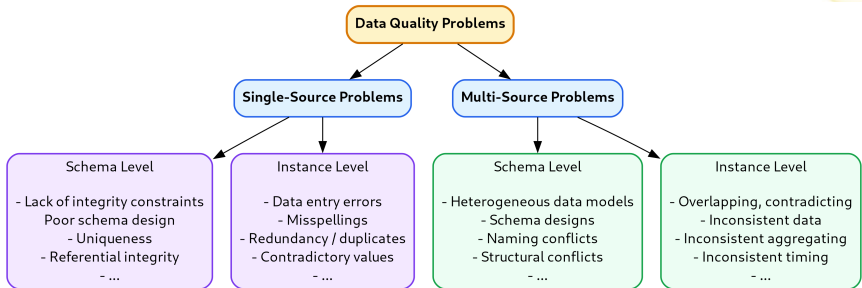
- **Supported by many languages**

- Pandas
- SQL GROUP BY operator
- Map-Reduce

```
In [94]: animals.groupby("kind").height.agg(
...:     min_height="min",
...:     max_height="max",
...: )
Out[94]:
```

	min_height	max_height
kind		
cat	9.1	9.5
dog	6.0	34.0

Classification of Data Problems



- Several problems **reduce trust and usability of data**
 - *Single-Source*: Occur within a single dataset
 - *Multi-Source*: Arise when integrating data from multiple sources
 - *Schema-Level*: Related to the structure or design of data
 - *Instance-Level*: Related to actual data values stored

Single-Source Problems

- **Depends largely on source**

- *Databases* often enforce schema / instance constraints
- *Spreadsheet data* is often “clean”
 - Schema exists
- *Logs* are messy
- *Web-page data* is messier

- **Types of problems**

- Ill-formatted data
- Missing/illegal values, misspellings, wrong fields, extraction issues
- Duplicated records, contradicting info, referential integrity violations
- Unclear default/missing values
- Evolving schemas/classification schemes (categorical attributes)
- Outliers

Single-Source Problems: Example

- **Sources of errors in data**

- Data entry errors: arbitrary values entered
- Measurement errors: sensor data inaccuracies
- Distillation errors: processing and summarization issues
- Data integration errors: inconsistencies across combined sources

Scope/Problem		Dirty Data	Reasons/Remarks
Attribute	Missing values	phone=9999-999999	unavailable values during data entry (dummy values or null)
	Misspellings	city="Liipzig"	usually typos, phonetic errors
	Cryptic values, Abbreviations	experience="B"; occupation="DB Prog."	
	Embedded values	name="J. Smith 12.02.70 New York"	multiple values entered in one attribute (e.g. in a free-form field)
	Misfielded values	city="Germany"	
Record	Violated attribute dependencies	city="Redmond", zip=77777	city and zip code should correspond
Record type	Word transpositions	name ₁ ="J. Smith", name ₂ ="Miller P."	usually in a free-form field
	Duplicated records	emp ₁ =(name="John Smith",...); emp ₂ =(name="J. Smith",...)	same employee represented twice due to some data entry errors
	Contradicting records	emp ₁ =(name="John Smith", bdate=12.02.70); emp ₂ =(name="John Smith", bdate=12.12.70)	the same real world entity is described by different values
Source	Wrong references	emp=(name="John Smith", deptno=17)	referenced department (17) is defined but wrong

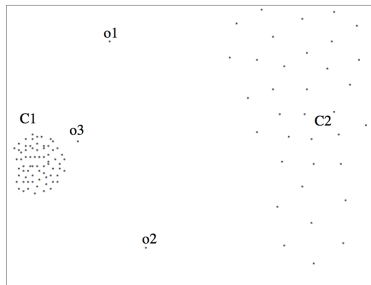
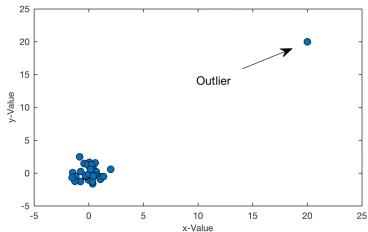
Table 2. Examples for single-source problems at instance level

Multi-Source Problems

- **Different data sources**
 - Developed separately
 - Maintained by different people
 - Stored in different systems
- **Schema mapping / transformation**
 - Map information across sources
 - Naming conflicts
 - Same name for different objects
 - Different names for same objects
 - Different representations across sources
- **Entity resolution**
 - Match entities across sources
- **Data quality issues**
 - Contradicting information
 - Mismatched information

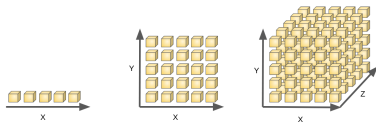
Univariate Outlier Detection

- **Problems with outliers**
 - Extreme outlier may alter metrics, masking others
- Use statistics to **identify outliers**
 - Robust statistics: minimize effect of corrupted data
 - Robust center metrics
 - Median
 - k%-trimmed mean (discard lowest and highest k% values)
 - Robust dispersion
 - Median absolute deviation (MAD)
 - Median distance from median value



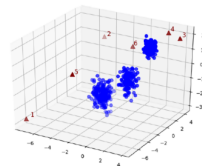
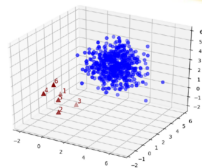
Outlier Detection

- **For Gaussian data**
 - Data points $1.5 \times$ MAD from median
 - Plot histogram to confirm
- **For non-Gaussian data**
 - Estimate generating distribution (parametric or not)
 - Distance-based: find points with few neighbors
 - Density-based:
 - Define *density* as average distance to k nearest neighbors
 - Use relative density to identify outliers
- **Curse of dimensionality**
 - “In high dimensional spaces, data is sparse”
 - Techniques break down as data dimensionality increases
 - Need $O(e^n)$ points with n dimensions to estimate
 - Project data into lower-dimensional space to find outliers
 - Not straightforward



Multivariate Outliers

- **Mean/covariance not robust**
 - Sensitive to outliers
 - Use robust statistics
- Assume **multivariate Gaussian data**
 - Defined by *mean* $\underline{\mu}$ and *covariance matrix* $\underline{\Sigma}$
 - *Iterative approach*
 - Mahalanobis distance: $\sqrt{(\underline{x} - \underline{\mu})^T \underline{\Sigma}^{-1} (\underline{x} - \underline{\mu})}$
 - Measures distance of point $\underline{x}$ from distribution
 - Identify outliers: points too far by Mahalanobis distance
 - Remove outliers
 - Recompute mean and covariance
- Data volume often large
 - Use approximation techniques
- Try different techniques based on data



Time Series Outliers

- Often data is in the form of a **time series**
- A **time series** is a sequence of data points recorded at regular intervals
 - Stock prices
 - Sales revenue
 - Website traffic
 - Inventory levels
 - Energy consumption
 - Market demand
 - Social media engagement
 - Hourly energy usage
 - Weekly retail foot traffic
 - ...
- Rich literature on **forecasting** in time series data
- Use historical patterns to flag **outliers**
 - Rolling MAD (median absolute variation)

